

Head-to-head Evaluation of Translation-First vs Agentic Generate→Validate→Repair Pipelines for Intent-Driven NetOps Changes — Validator-Acceptance Parity under Shared-Candidate Semantics

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Abstract—We report a fully preregistered, artifact-archived paired experiment (CAND-2026-001-prereg-v1) that compared two operational architectures for intent→configuration NetOps changes across heterogeneous vendor CLIs and Mininet-derived topologies: (A) translation-first (constrained vendor DSL → formal verifier → solver) and (B) agentic generate→validate→repair (hosted LLM + deterministic validators). The run declared gpt-5-mini for generation and deterministic_netops_validator_v5 for deterministic end-to-end correctness checks; preregistration used shared_initial_candidate semantics so each task pair received a single LLM candidate shared between arms. Execution completed 240 episodes (120 paired tasks) with model_calls_used = 120 (one shared generation per pair). Deterministic scoring produced contingency counts n11 = 120, n10 = 0, n01 = 0, n00 = 0 (baseline = 120/120; guarded = 120/120). Paired difference (A - B) = 0.00; paired-bootstrap 95% CI = [0.00, 0.00] (10,000 resamples, seed 1729); exact McNemar two-sided p = 1.0. Because generation was intentionally shared, these outcomes measure validator-acceptance parity for identical candidates rather than independent generator or repair robustness. We provide artifact-grounded evidence (execution and analysis manifests, per-task validator traces), an explicit evidence-to-claim mapping table, a nearest-work comparison matrix, and preregistered protocol modifications required to obtain a discriminating head-to-head comparison in follow-ups.

I. INTRODUCTION

Automating intent→configuration translation and safe sequencing of multi-vendor NetOps changes is operationally critical: production networks require both correctness guarantees (reachability, isolation, absence of transient safety violations) and flexibility to express diverse intents across vendor CLIs. Two architectural approaches are common: translation-first pipelines that restrict generation to a vendor DSL and apply formal verification/solvers, and agentic generate→validate→repair pipelines that centralize generation in an LLM and rely on deterministic validators plus bounded repair. Prior literature emphasizes both deterministic guarantees (formal verification) and the promise of LLM-driven flexibility, but direct, preregistered comparisons that produce artifactized per-task validator traces are rare.

This study’s preregistered head-to-head question was: for intent-driven network changes across heterogeneous vendor

CLIs and realistic emulator topologies, which architecture yields higher end-to-end operational correctness (measured by deterministic validation: reachability and policy isolation) — (A) translation-first or (B) agentic generate→validate→repair? The preregistration (CAND-2026-001-prereg-v1) fixed a paired design (N = 120 paired tasks), the primary estimand (paired difference in binary end-to-end correctness A - B), the generator (gpt-5-mini), the deterministic validator (deterministic_netops_validator_v5), and the analysis executor (paired_binary_analysis_v1).

A decisive preregistered design choice was shared_initial_candidate semantics: each paired task used a single LLM candidate generated once and supplied identically to both architectures. This choice deliberately removes generator variance from the confirmatory estimand, turning the experiment into a validator-acceptance parity test for identical candidates rather than an independent generator or repair efficacy comparison. The remainder of the paper (Methods, Results, Discussion) makes that restriction explicit and maps preregistered hypotheses to the measured estimand with artifact references.

II. RELATED WORK

Two research traditions frame our contribution and motivate the experimental axes used in this study: (1) formal translation and verification systems that prioritize deterministic guarantees, and (2) LLM-driven agentic NetOps work that prioritizes flexible generation and iterative repair. Below we expand these threads and contrast their experimental semantics, instrumentation, and typical evaluative practices—focusing specifically on the axes used by our preregistered design (independent generation sampling, repair-loop instrumentation, validator acceptance focus, and emulator/hardware realism).

Formal translation→verification. A long line of work treats network configuration synthesis and verification as a constrained-synthesis problem: generate configurations in a restricted DSL, then apply solver-based or symbolic verification to obtain soundness guarantees for reachability and isolation. Seminal systems such as SymNet and follow-on "Don’t

Mind the Gap" emphasize determinism and programmatic representations of forwarding behavior; they enable localized, incremental checks and provide formal accounts of when synthesized configuration changes preserve invariants [10,11]. More recent programming-language and verification advances (e.g., Weighted NetKAT and related quantitative verification work) extend the expressiveness of DSLs and enable richer objectives (cost, weighted properties) while retaining verifier semantics that produce binary accept/reject decisions or counterexamples suitable for solver feedback [5]. The experimental semantics common to these systems are: (i) a constrained generation surface (DSL) that bounds output space; (ii) formal validators that operate on an abstract semantics of device behavior; and (iii) low generator variance because synthesis operates inside the DSL or under solver guidance. Empirical comparisons in this family typically focus on solver scalability, counterexample diagnostics, and incremental verification throughput rather than stochastic generator variance.

LLM-driven and agentic NetOps. In contrast, the recent wave of LLM-powered NetOps and AIOps research treats translation as a flexible natural-language→CLI mapping problem and emphasizes sample-based generation plus iterative repair. Representative recent works explicitly instrument generate→validate→repair pipelines and multi-agent verifier loops to reduce hallucination and unsafe proposals [3,4]. These LLM-centric evaluations commonly exercise independent generation sampling (multiple model calls per condition), record repair iteration traces, and measure repair coverage and transient unsafe-proposal rates as operational metrics. Because LLM outputs are stochastic and can manifest structural errors, LLM-centric experiments often allocate larger model-call budgets, instrument per-generation diagnostics (tokens, parse errors), and adopt deterministic validators as downstream safety gates. Critically, LLM experiments that compare pipelines frequently differ in whether they (a) compare independent generator draws per arm (which measures generator+orchestrator combined performance) or (b) hold the generator output fixed to contrast downstream orchestration and verification behaviors.

Experimental axes and their methodological consequences. Prior work therefore occupies distinct points along the four experimental axes we emphasize: independent generation sampling, repair-loop instrumentation, validator-acceptance focus, and emulator/hardware realism. Translation-first systems score highly on deterministic guarantees and low generator variance (because synthesis is constrained), while LLM-agent experiments emphasize generator sampling, repair coverage, and stochastic robustness. Mahmood & Song (IFIP Networking 2026) report a self-correcting multi-agent verifier pipeline instrumenting repair loops and acceptance thresholds; INTA (ICNP 2025) and related intent→configuration studies evaluate translation strategies and LLM agents with independent generation sampling to capture generator variance [3,4]. Weighted NetKAT and related formal frameworks supply the deterministic foundation and are commonly used as the downstream verifier in translation-first evaluations [5]. These verified works

are the direct methodological relatives of the present run.

Positioning of the present run. The distinguishing experimental decision in our preregistration is shared_initial_candidate semantics: for each paired task we generated a single LLM candidate and supplied that identical candidate to both the translation-first and agentic arms. Methodologically, this places our run orthogonally to independent-generation LLM experiments: rather than estimate generator variance or repair coverage, the run isolates differences in downstream acceptance and orchestration given identical candidate proposals. This choice mirrors verifier-parity regression checks frequently used when deploying new validators or orchestration stacks (i.e., when operators need to know whether swapping validators/orchestrators changes acceptance outcomes for the same proposals). Nearest works that instrument repair loops and sample independently remain complementary: to discriminate generator robustness or repair efficacy one must change the generation semantics (independent per-arm draws and nonzero repair budgets) and correspondingly increase model_call budgets and repair instrumentation. Our discussion and preregistered recommended protocol modifications make these resource and instrumentation differences explicit.

Artifact and evaluation practices across prior work. A practical contrast concerns archival and per-task artifactization. Formal translation studies typically archive DSL programs, constraints, and counterexamples produced by solvers; LLM agent studies archive multiple model responses, repair prompts/traces, and validator logs. The present run follows the LLM-artifact practice but intentionally reduces generator artifact multiplicity by using one shared response per pair; archival traces therefore emphasize per-task validator snapshots (validation_before/after), repair_applied flags, and the deterministic acceptance decision. This archival focus enables reproducible validator-parity claims while preserving the ability to modify generation semantics in follow-ups to test the broader preregistered hypotheses about independent generator and repair differences.

III. METHODOLOGY

Preregistration and execution contract. The study followed the machine-readable preregistration CAND-2026-001-prereg-v1, which fixed the confirmatory design (paired binary estimand A - B), sampling ($N_{\text{confirmatory}} = 120$ paired tasks), generator (declared_model_version = gpt-5-mini), deterministic validator (deterministic_netops_validator_v5, validator_version = 5.0), adapter_family = hosted_netops_gvr_v1, and analysis executor (paired_binary_analysis_v1). Key execution limits were recorded in the preregistration and enforced by orchestration: maximum_model_calls = 240, planned_episode_count = 240, initial_generation_calls_per_task = 1, and maximum_repair_calls_per_task = 1. The preregistration also specified the analysis procedure (bootstrap percentile 95% CI with 10,000 resamples and bootstrap_seed = 1729; exact McNemar two-sided test) and the primary

exclusion, missingness, and contamination rules. The archived preregistration is the authoritative design freeze for the run.

Task generation, balancing, and calibration. Confirmatory tasks were produced deterministically by `deterministic_netops_task_generator_v5`. The task manifest entries (`execution/task_manifest.jsonl`) include: intent text, canonical reference answer, initial and expected final states, a typed `required_sequence` (ordered field/interface/value changes), `allowed_touched_fields`, `workflow_policies` (e.g., `make-before-break`, `all_down_before_any_field_change`), and a compact difficulty vector (`changed_interface_count`, `dependency_rule_count`, `required_command_count`, `level` $\in$ {medium, hard, very_hard}). Tasks were balanced across preplanned vendor/topology clusters per the preregistration distribution (3 vendors $\times$ 3 topology classes, assigned across clusters to total 120). The deterministic task generator and these metadata fields enable the validator to perform exact, intention-preserving checks rather than approximate semantic matching.

Shared_initial_candidate semantics and orchestration accounting. The orchestration implemented the preregistered `shared_initial_candidate` stage: for each pair the automation performed a single hosted LLM call that produced one textual candidate; that candidate was then supplied identically to both downstream pipeline arms (translation-first baseline and agentic guarded orchestration). Execution accounting is archived (`execution/execution_manifest.json`) and shows `planned_episode_count` = 240, `completed_episode_count` = 240, `failed_episode_count` = 0, and `model_calls_used` = 120 — consistent with one shared generation per pair. Provider audit recorded `hosted_model_call_event_count` = 120 and `cross_condition_cache_key_reuse_observed` = false, demonstrating fresh, isolated model contexts per shared call as prescribed in the `contamination_plan`.

Validator design and deterministic scoring rules. The `deterministic_netops_validator_v5` executes a multi-stage `full_intent_constraint_validation` pipeline applied to each candidate. Pipeline stages are: (1) syntactic normalization and token-level parsing into commands/fields; (2) required-command detection (compare parsed commands against `task.required_sequence` and `allowed_touched_fields`); (3) dependency and ordering verification (enforce `make-before-break`, `shutdown`→`modify`→`restore` patterns); (4) simulation of deterministic final state effects in the Mininet-derived emulator nominal model and check of reachability/isolation invariants; (5) comparison to `expected_final_state`; (6) scoring output assembly (`validation_before/validation_after` snapshots, `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, `violation_codes` list). The validator emits `validation_after.valid` (boolean) and `score_reason_code` (e.g., `VALID_CONFIGURATION`) and a `repair_applied` flag when a repair step was actually executed by a downstream repair actor. A pair is scored success (1) only when `validation_after.valid == true`

under the `full_intent_constraint_validation` rule set. All per-task validator traces and scorer outputs were archived (`execution/scoring/task-000XXX-*.json`) and are the canonical evidence used by the analysis executor.

Repair policy, bounded repair budget, and retry handling. The agentic pipeline design included a bounded repair loop: `maximum_repair_calls_per_task` = 1 (preregistered). Repair behavior was conditioned on validator failures; when repair iterations were permitted the orchestration would generate a repair prompt that included the validator trace and a request to minimally modify the candidate to satisfy the validator. Repair prompts, repair responses, and `repair_applied` indicators were recorded in the `repair_provider_trace_path` fields of the scoring JSONs. The execution contract also specified API-level retry policies for infrastructure resilience: transient hosted-model/API outages were retried twice with exponential backoff (1s then 3s); nonrecoverable provider errors or exhausted retries were handled per `missingness_plan` and logged in the execution manifest; these machine-recorded events form the reproducible audit trail.

Contamination, fidelity checks, and exclusion rules. The preregistered `contamination_plan` mandated two classes of preflight diagnostics: (i) simulator fidelity checks using a held-out deterministic suite of 50 vendor examples to measure simulator↔public-device disagreement, with an exclusion threshold of >5% disagreement; and (ii) vendor grammar coverage/unit tests to detect parser unknown constructs. Orchestration logged contamination checks and produced `analysis/contamination_summary.csv`; no contamination flags were raised for this run. Exclusions required by the preregistration would have been applied and reported, but none occurred: `analysis/missingness_summary.csv` shows `complete_pairs` = 120 and no failed episodes.

Analysis pipeline and reproducible parameters. Analysis followed the archived `paired_binary_analysis_v1` executor. For each confirmatory pair the executor consumes per-task scoring JSONs, computes the binary end-to-end correctness indicators for baseline and guarded, constructs the 2×2 contingency (`n11,n10,n01,n00`), computes the paired difference (`A - B`), and produces the paired bootstrap percentile 95% CI (`bootstrap_resamples` = 10000, `bootstrap_seed` = 1729). The executor also runs an exact McNemar two-sided test for paired binary outcomes. All analysis outputs and parameters are archived (`analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/analysis_log.jsonl`, `analysis/deterministic_reconciliation.json`) to permit deterministic recomputation. Secondary estimands (repair coverage, mean repair iterations) were implemented in analysis but are only informative when `repair_applied` flags are nonzero; the analysis codepath and decision logic are included in the archived executor so null or zero counts are reproducibly verifiable.

Reproducibility posture. The methodology enforces deterministic generators for the confirmatory manifest, a fixed LLM identifier (`declared_model_version` = `gpt-5-mini`) for the shared generation stage, a single deterministic validator version (`deterministic_netops_validator_v5 v5.0`), and an

archived analysis executor with explicit random seeds. Per-task artifacts (responses and scoring JSONs), execution manifests, and analysis artifacts are available in the execution bundle referenced by the execution manifest; these artifacts contain the exact inputs, outputs, validator traces, and analysis parameters required to reproduce the reported estimands without changing the preregistered design.

IV. RESULTS

Execution accounting and primary numeric outcomes. The execution manifest records `planned_episode_count = 240`, `completed_episode_count = 240`, `failed_episode_count = 0`, `model_calls_used = 120` (one shared LLM call per paired task), and `maximum_model_calls = 240`. Deterministic reconciliation (`analysis/deterministic_reconciliation.json`) confirms `complete_pair_count = 120` and provider audit counts: `hosted_model_call_event_count = 120`, `completed_hosted_model_call_event_count = 120`, `failed_hosted_model_call_event_count = 0`, `cache_miss_count = 120`, `cache_hit_count = 0`, and `cross_condition_cache_key_reuse_observed = false`. These provider-audit numbers substantiate that the orchestration created exactly one independent generation per pair and maintained execution isolation as preregistered.

Primary contingency and statistical summaries. The analysis contingency (`analysis/paired_contingency_table.csv`) is `n11 = 120`, `n10 = 0`, `n01 = 0`, `n00 = 0`. Marginal success rates are `baseline = 120/120 = 1.0` and `guarded = 120/120 = 1.0`; the observed paired difference `A - B = 0.00`. The paired bootstrap percentile 95% CI was computed with `bootstrap_resamples = 10,000` and `bootstrap_seed = 1729` (`analysis/analysis_log.jsonl`) and returned `[0.00, 0.00]`. The exact McNemar two-sided test yields `p = 1.0` (test applied to the observed discordant counts `n10` and `n01`). These statistics are deterministic given the degenerate contingency table.

Repair and secondary outcomes. Across all 240 episodes the archived per-task scoring traces record `repair_applied = false` (see `execution/scoring/*.json`). Aggregation yields `repair_applied count = 0`; consequently preregistered secondary estimands that quantify repair coverage and mean repair iterations are unobserved in this execution. The absence of repair activity is present in every representative scoring JSON (for example, `execution/scoring/task-000001-baseline.json` records `repair_applied: false` and `validator_trace.validation_after.valid: true`), and is consistent across the artifact set.

Representative per-task diagnostics. Representative scoring artifacts show that `shared_initial_candidate` matched the `normalized_reference_answer` modulo command ordering in several cases and that validators parsed and validated required commands: example excerpts from archived scoring files illustrate typical complexity and validator checks—

- `task-000001`: `parsed_command_count = 4`, `required_command_count = 4`, `validation_after.valid = true`; `normalized_configuration` and `final_state` fields show the expected interface `admin/mtu/vlan` values

and no violations (`execution/scoring/task-000001-*.json`).

- `task-000002`: `parsed_command_count = 2`, `required_command_count = 2`, `validation_after.valid = true`; difficulty flagged as `level = hard` with `pattern = make_before_break_uplink_switchover`.
- `task-000003`: `parsed_command_count = 6`, `required_command_count = 6`, `validation_after.valid = true`; difficulty = `hard` with `pattern = paired_link_atomic_mtu_change`.

These representative entries demonstrate that the validator applied dependency and ordering checks (`make-before-break`, `shutdown→modify→restore`) and that the normalized responses satisfied those constraints for the shared candidates.

Contamination and missingness diagnostics. The preregistered contamination checks produced no flags: `analysis/contamination_summary.csv` is empty. Missingness diagnostics (`analysis/missingness_summary.csv`) report `complete_pairs = 120` and zero for `baseline_only_observed_pairs`, `guarded_only_observed_pairs`, `both_missing_pairs`, and `failed_episodes`. Provider audit and execution logs show no retries or infrastructure failures beyond the planned single generation per pair; `execution/execution_manifest.json` and `analysis/deterministic_reconciliation.json` contain the machine-readable accounting.

Interpretation constrained to measured estimand. Because the design used `shared_initial_candidate` semantics and a single deterministic validator per arm, the numerical result—perfect concordance across all 120 paired tasks—directly measures validator-acceptance parity for identical LLM candidates. The degenerate contingency produces a point mass estimate (difference = 0.00) and a degenerate bootstrap CI and p-value; these are accurate descriptions of the observed data. However, by design these results do not address independent generator sampling variance, multi-sample selection, or the effectiveness of repair loops when the generator is called separately for each arm. Those secondary dimensions remained unexercised here and are therefore unobserved in the artifacts.

V. DISCUSSION

What the run measures — and what it does not. Because `shared_initial_candidate` semantics were used, the preregistered confirmatory estimand reduces to validator-acceptance parity: given identical LLM candidates, do downstream orchestration and verification paths differ in deterministic validator acceptance? The observed complete concordance (`n11 = 120`) answers this restricted question: both validators accepted the identical candidates for every paired task. The run does not exercise independent generator variance, sampling effects, or repair efficacy; therefore claims about which architecture would produce more correct initial candidates or achieve higher repair coverage when generating independently are untested.

Evidence-to-claim mapping (compact table). Table 2 (in manuscript) maps preregistered hypotheses/estimands to measured observables and archived artifacts. Example mappings: `H1 (paired difference A - B) → measured estimand = validator-acceptance parity`

under shared_initial_candidate; supporting artifacts: analysis/paired_contingency_table.csv and analysis/condition_summary.csv (n11=120,n10=0,n01=0,n00=0); verdict: tested but restricted. H2 (agentic repair coverage advantage) → observed repair_applied count = 0 in execution/scoring/*.json; verdict: unobserved in this execution.

Operational implications and protocol guidance for discriminating comparisons. The artifact bundle documents a reproducible protocol for validator regression and orchestrator equivalence checks under fixed candidates. To test independent-generation and repair differences the pre-registered protocol must be modified (and these modifications can be preregistered and audited): (i) remove shared_initial_candidate so each arm receives an independent LLM generation per pair (model_calls_used would rise from 120 to 240 for initial candidates); (ii) allow multiple sampled initial generations per arm to estimate generator variance; (iii) enable and instrument repair iterations so repair_applied counters become informative; (iv) increase emulator fidelity or include hardware runs to reduce simulator-to-hardware contamination risk. Each modification yields verifiable changes in execution artifacts (model_calls_used, hosted_model_call_event_count, nonzero repair_applied flags) and therefore supports reproducible discrimination across architectures.

Threats to validity (concise). Internal validity: shared_candidate removes generator variance by design; this is a feature not a flaw but narrows the estimand. Construct validity: deterministic validator acceptance operationalizes a conservative correctness definition but omits human-facing properties (explainability, maintainability). External validity: synthetic tasks and Mininet-derived emulation plus a single model (gpt-5-mini) and single validator (deterministic_netops_validator_v5) limit generality. Conclusion validity: degenerate concordance (all pairs concordant) produces point mass CI and $p = 1.0$; these statistics correctly describe the observed data but do not imply production-level equivalence under independent operation.

Reproducibility and audit artifacts. All primary artifacts are archived and referenced by execution/execution_manifest.json: preregistration (CAND-2026-001-prereg-v1), execution/model_configuration.json (declared_model_version = gpt-5-mini), execution/task_manifest.jsonl, execution/responses/*, execution/scoring/*.json, analysis/paired_contingency_table.csv, analysis/condition_summary.csv, analysis/analysis_log.jsonl, and analysis/deterministic_reconciliation.json. Per-task validator traces (execution/scoring/task-000XXX-*.json) are the canonical records for each scored decision. The analysis executor parameters (bootstrap_resamples = 10000, bootstrap_seed = 1729) are recorded in analysis/analysis_log.jsonl and enable deterministic recomputation of the reported CI.

VI. LIMITATIONS

- Preregistered shared_initial_candidate semantics were used; this intentionally removes generator variance and prevents this run from assessing independent generation robustness differences between architectures.
- The task corpus was synthetic and executed in an emulator (Mininet-derived topologies and public vendor example grammars); emulator-to-hardware divergence limits external validity to production devices.
- A single LLM (gpt-5-mini) and a single deterministic validator (deterministic_netops_validator_v5) were used; results do not imply multi-model or multi-validator generality.
- No repairs were applied in this run (repair_applied = false in per-task traces); preregistered secondary estimands about repair coverage remain unobserved for this execution.
- The binary deterministic validator acceptance metric does not capture operator-facing properties such as explanation quality, maintainability, or human trust, which matter in production but were outside the metric used here.

VII. CONCLUSION

We executed a preregistered, artifactized paired experiment comparing translation-first and agentic generate→validate→repair NetOps pipelines under shared_initial_candidate semantics. For 120 paired tasks deterministic validators accepted identical LLM candidates in both arms for every pair (n11 = 120, n10 = n01 = n00 = 0), yielding paired difference 0.00 (bootstrap 95% CI [0.00, 0.00], McNemar $p = 1.0$). No repairs were applied (repair_applied = false for all episodes); preregistered hypotheses about repair coverage are therefore unobserved. The run precisely measures validator-acceptance parity under controlled shared candidates and provides a reproducible artifact bundle and explicit protocol modifications for future discriminating comparisons that exercise independent generation and repair coverage.

DISCLOSURE STATEMENT

This research was executed autonomously after the autonomy lock under the frozen master prompt archived with the run provenance. Preregistration: CAND-2026-001-prereg-v1 (archived and used to freeze the design; preregistration_sha256 recorded in the execution manifest). Generation used gpt-5-mini (declared_model_version in execution/model_configuration.json); provider record 'openai_responses'. The hosted adapter family was hosted_netops_gvr_v1. Deterministic artifacts included deterministic_netops_task_generator_v5 and deterministic_netops_validator_v5 (validator_version 5.0). The run deliberately used the preregistered shared_initial_candidate generation semantics: both pipeline arms received the same initial LLM candidate per task; execution accounting shows model_calls_used = 120 (one shared generation per pair) while planned episodes were

240. Primary analysis used paired bootstrap percentile CIs (10,000 resamples, seed 1729) and an exact McNemar two-sided test executed by `paired_binary_analysis_v1`. No human manually edited, rewrote, or post-processed technical content after the autonomy lock. The Research Director selected the broad topic family and constrained the initial master prompt prior to the autonomy lock; the autonomous run performed literature review, final research-question selection, preregistration, code generation, experiment execution, analysis, peer review, and manuscript drafting after the lock. Immutable reference to the initial master prompt: `provenance/master_prompt.txt` (SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba). The full execution and analysis artifact bundle is archived and referenced by the execution manifest included with this submission.

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